Overfitting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit the leaves. Compare new cases. Investigate the gap.

Distinguish inspectable training-specific fit from an unsupported universal capacity diagnosis.

Keep one habit: Supplied MAEcomparison → pictured accuracy gaps → complete depth0/1/2paper trees → independent new results.

MAIN LESSON · SECTION 1

Did the extra flexibility improve new-case evidence?

> Perfect training fit is not the objective or a complete diagnosis.

Learning goals

I can compare **training** and validation **evidence** for model **capacity**.

I can explain **overfitting** without assuming every gap proves it.

I can use a declared selection rule instead of perfect **training** fit.

Recall training and validation roles

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.
- Can a zero **training** error guarantee good validation performance?

End goal

By the end, I can explain **overfitting** using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

An overfit model captures **training**-specific noise, producing a large gap between **training** and unseen performance.
</details>



LEARNING TARGET A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Compare capacity under the same metric and source rules

> Inspect leakage,coverage and uncertainty before attributing a gap to model size.

Capacity explanation and fictional scope

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **overfitting** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Read all training and validation results together

> The lowest training number can coexist with the worst validation result.

Complete supplied three-candidate table

Main supplied regression candidate	Training MAE	Validation MAE	Validation minus training loss
Constant	5	6	1
Line	2	3	1
Flexible curve	0	8	8

On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Explain selection,gap direction and evidence scope

The declared selection rule is smallest validation MAE, so the line wins this supplied comparison. These are aggregate results, not the training records or implementation of three actual fitted programs. A loss gap is validation minus training here; the classification picture uses training minus validation accuracy, where larger accuracy is better. State which direction and unit you are using.

The complete paper-tree lab below supplies separate case-level fitting evidence.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

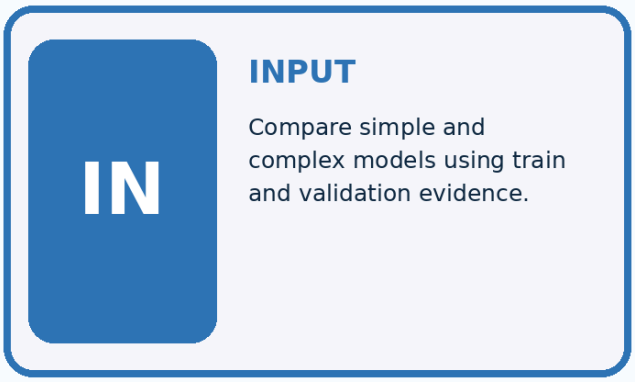
Keep the distinction between an observed result and a proposed next step.

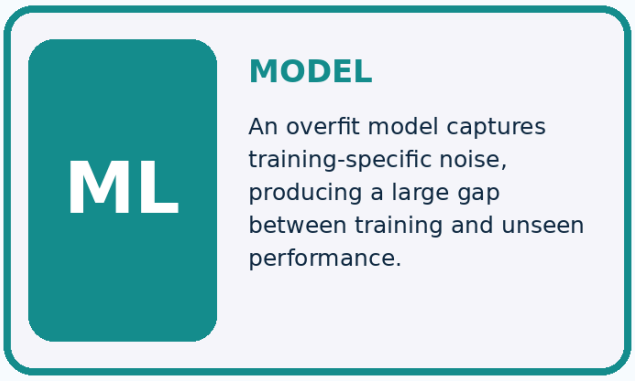
▼ Earlier wording, alternatives and picture transcripts

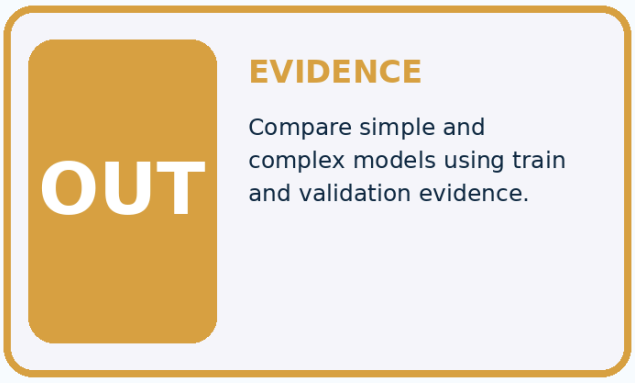
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Compare simple and complex models using train and validation **evidence**.

MODEL IDEA An overfit model captures **training**-specific noise, producing a large gap between **training** and unseen performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Simple tree: train 82%, validation 78%; gap 4 points. Deep tree: train 100%, validation 65%; gap 35 points.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

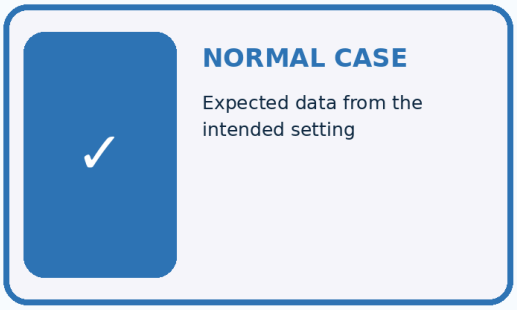
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

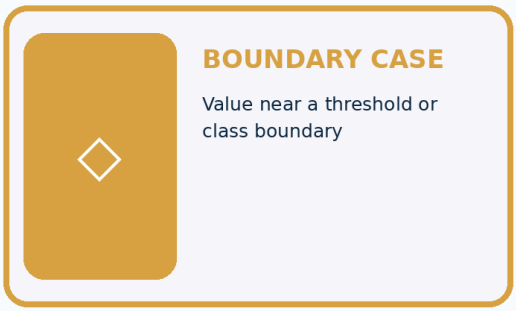
Compare a capacity hypothesis with the evidence for it

> A visible gap needs a source and sampling audit as well as a model choice.

Four evaluation conditions



NORMAL Expected data and expected behavior.

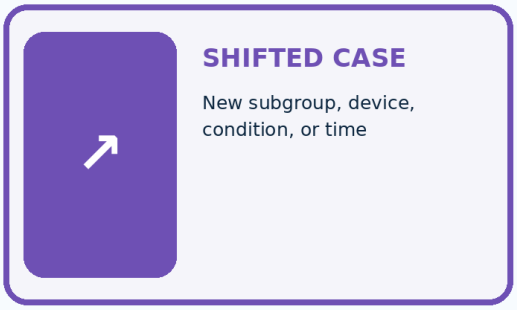


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Large’s validation accuracy rises to82% after a development revision.

▼ **Check after predicting**

Prediction: Its gap is 18 percentage points; it now beats Small on this validation set.

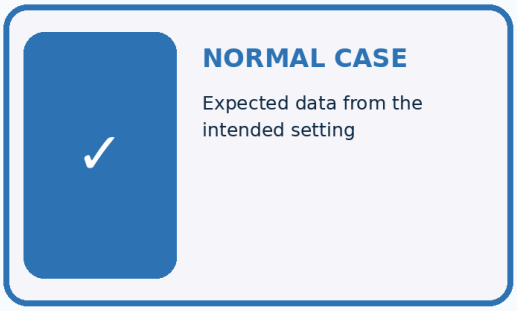
Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

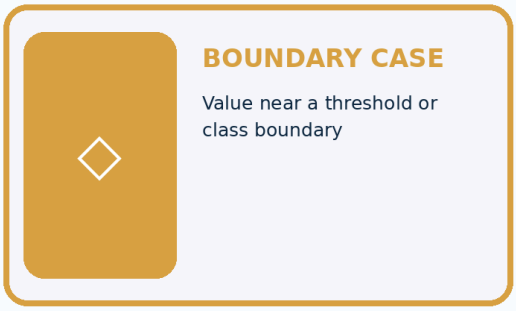
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

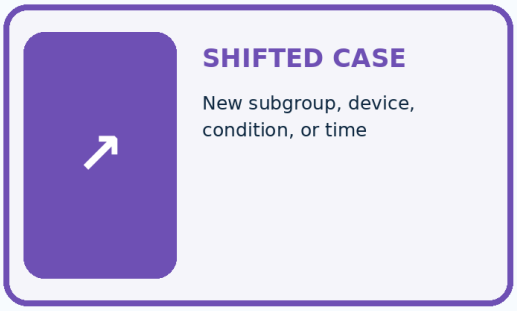
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace gap and validation score as two distinct quantities

> A smaller gap need not identify the best development candidate.

Complete pictured results and separate-fixture scope

See **overfitting** as a gap requiring investigation, not a label based only on model size.

Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

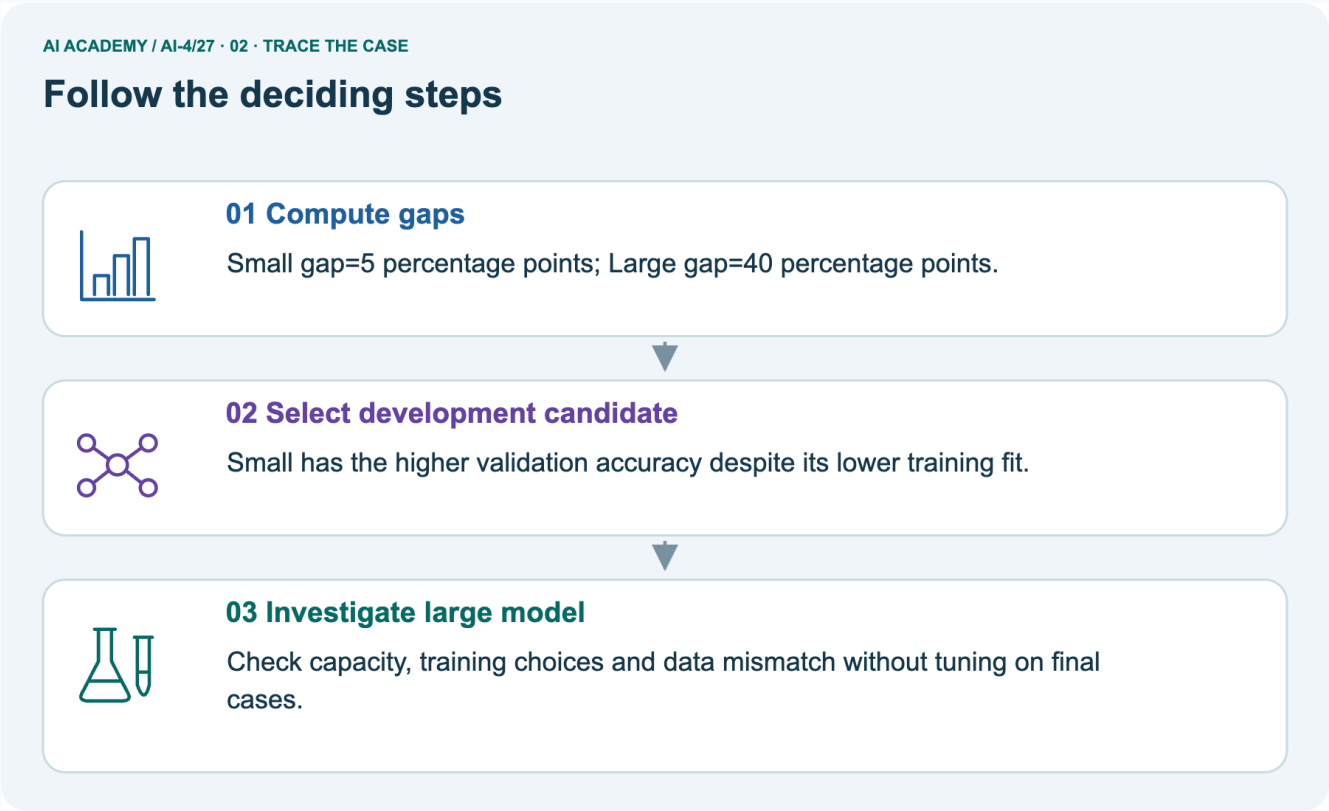
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Picture Small has training 85%, validation 80%, gap 5 percentage points; Large100%/60%, gap 40 points. The pictured revision changes Large's validation to82%while leaving training 100%, so its gap 18 points remains larger yet its validation score exceeds Small's80%. Gap

size alone is not the selection rule. These supplied accuracy results are separate from the main MAEtable and the paper-tree 4-case splits.

No sample-size uncertainty, causal capacity effect or final-test score is supplied.

Trace gaps and candidate selection



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

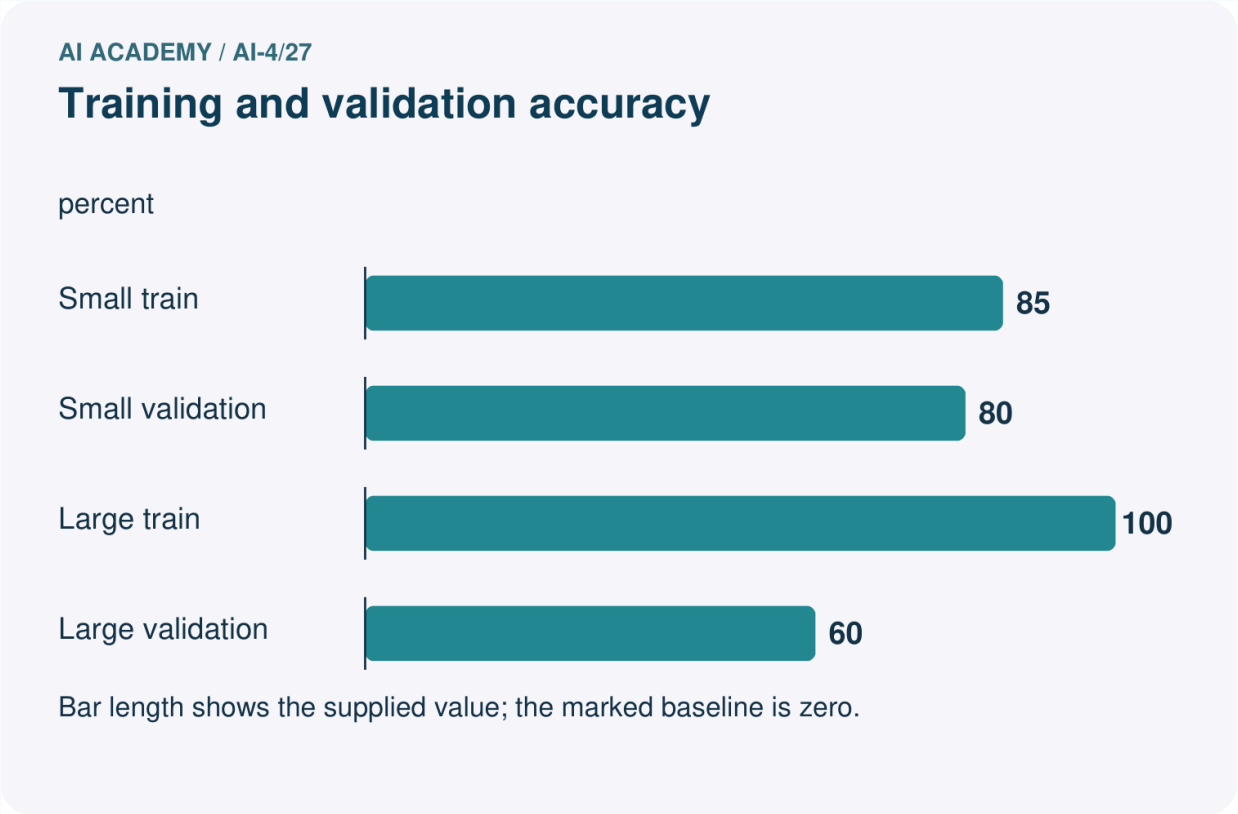
STEP	WHAT HAPPENS AND WHY
1. Compute gaps	Small gap=5 percentage points; Large gap=40 percentage points.
2. Select development candidate	Small has the higher validation accuracy despite its lower training fit.
3. Investigate large model	Check capacity, training choices and data mismatch without tuning on final cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original train/validation comparison graphic

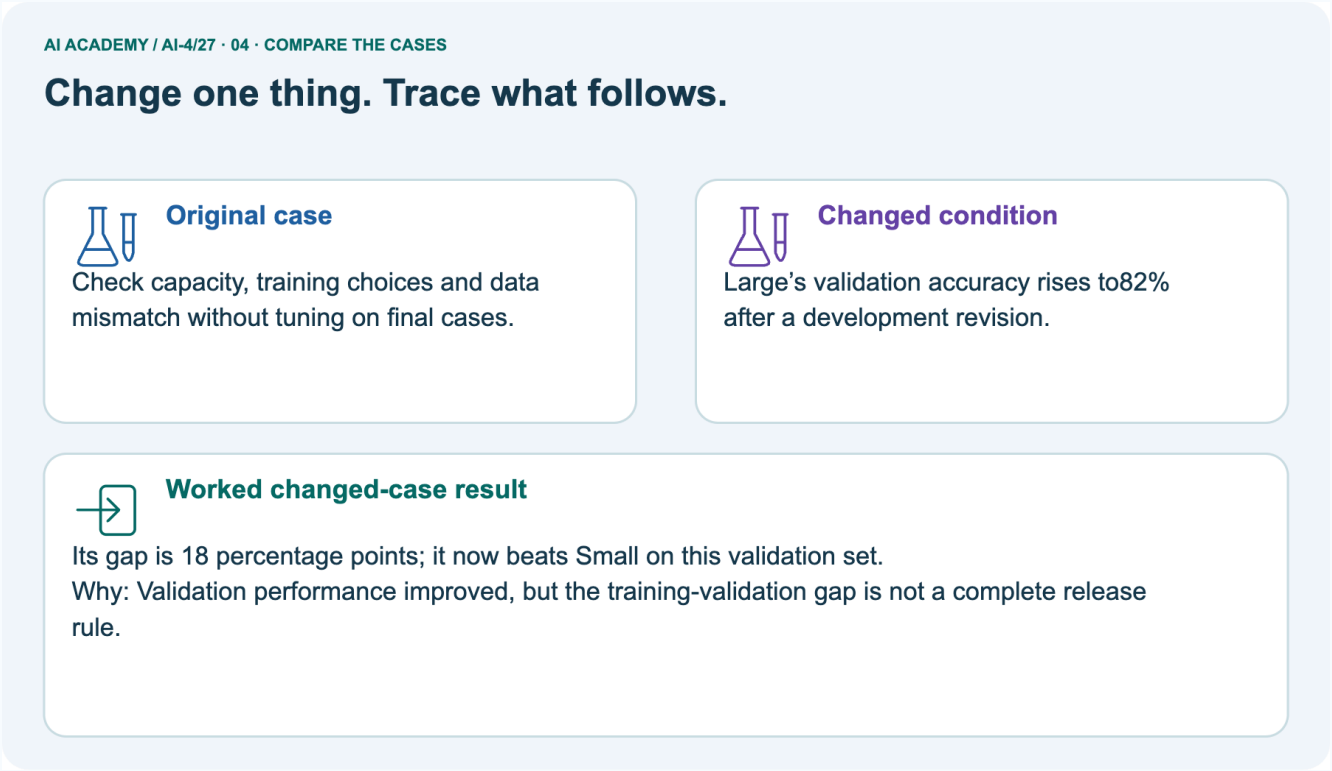


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict the revised82%validation result

- Change one thing: Large’s validation accuracy rises to82% after a development revision. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain alternative causes and selection limits

- A gap can also reflect data shift or sampling noise. Inspect **evidence** before attributing every gap to **capacity**.

- Read the labelled picture: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main gap-diagnosis question

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 27 • WORKED EVIDENCE

1. Simple tree: train 82%, validation 78%; gap 4 points.
2. Deep tree: train 100%, validation 65%; gap 35 points.
3. Deep tree's perfect training result is weaker unseen evidence.
4. Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

STEP 1 Simple tree: train 82%, validation 78%; gap 4 points.

STEP 2 Deep tree: train 100%, validation 65%; gap 35 points.

STEP 3 Deep tree's perfect **training** result is weaker unseen **evidence**.

STEP 4 Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Overfitting

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find perfect training fit mistaken for generalization

> A development result can suggest an issue without isolating its cause.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The model with the highest **training** accuracy is the best model.

Explain the limitation and next check

- A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

  Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

> **What extra evidence would you inspect before concluding a model memorized noise?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model with the highest training accuracy is the best model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise one recorded development change at a time

> Retain difficult valid cases and protect the final evaluation.

Try supported W2

Use workbook W2 for the connected example “**Overfitting** and model **capacity**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Plan simplification,coverage or a declared setting change

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared **regularization** setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Explain the evidence beyond a final choice

- Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured training-perfect candidate

- Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent96/62and88/79task

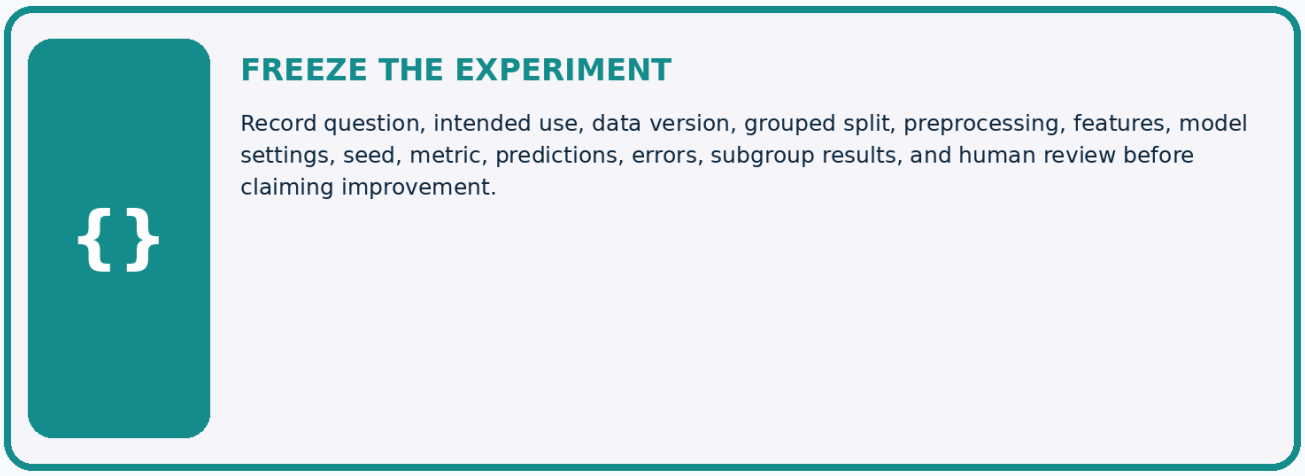
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Interpret a gap

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Complexity Trial with every fitting and validation card supplied

> The constrained paper method makes the training-specific fit inspectable.

Complete eight-case source table

Unique fictional trip ID · Role · Near flag · Tall flag · Reviewed reference

> **Unique fictional trip ID:** T1
>
> **Role:** Training
>
> **Near flag:** 0
>
> **Tall flag:** 0
>
> **Reviewed reference:** clear

> **Unique fictional trip ID:** T2
>
> **Role:** Training
>
> **Near flag:** 0
>
> **Tall flag:** 1
>
> **Reviewed reference:** clear

> **Unique fictional trip ID:** T3

Role: Training

Near flag: 1

Tall flag: 0

Reviewed reference: blocked

Unique fictional trip ID: T4

Role: Training

Near flag: 1

Tall flag: 1

Reviewed reference: clear

Unique fictional trip ID: V1

Role: Validation

Near flag: 0

Tall flag: 0

Reviewed reference: clear

Unique fictional trip ID: V2

Role: Validation

Near flag: 0

Tall flag: 1

Reviewed reference: clear

Unique fictional trip ID: V3

Role: Validation

Near flag: 1

Tall flag: 0

Reviewed reference: clear

Unique fictional trip ID: V4

> **Role:** Validation
>
> **Near flag:** 1
>
> **Tall flag:** 1
>
> **Reviewed reference:** blocked

Declared fitting,stopping and selection rules

Complexity Trial uses the complete eight-record table. Human-declare near as the first question and tall as the second; fit each reached leaf's training majority, choosing clear for label ties. Depth0is one training-majority constant; depth 1 may ask near; depth 2 may also ask tall inside a branch. Stop early if a branch's labels are identical. The question order is constrained, not searched for optimality.

Keep all four training and all four validation records fixed. Choose by largest validation accuracy, then shallower depth on a tie.

Fit,plot and explain the comparison

1. Build the depth 0/1/2 paper trees from the supplied training cards, derive each leaf label, and record every prediction on the unchanged validation cards. Plot depth horizontally and separate training/validation accuracy vertically with 4 cases per split. Try before opening the checked summary or running the optional fitter.

   Identify the training-improves/validation-worsens pattern, then name other data explanations still needing inspection. No fixed depth guarantees generalization and no final test is supplied. Keep the independent new accuracy comparison for an uncoached first attempt.

▼ **Check all leaf/score results after trying**

Maximum question depth	Training correct /4	Validation correct /4	Checked evidence
0	3 /4	3 /4	Predict clear everywhere.

Maximum question depth	Training correct /4	Validation correct /4	Checked evidence
1	3 /4	3 /4	Near0 leafclear; near 1 labelsblocked/cleartie, so leafclear.
2	4 /4	2 /4	Near1 then tall 0 predictsblocked, tall 1 clear; validationV3/V4 both fail.
Selection and limit	Trainingperfectat depth2	Choose depth 0 under validation/simplicity rule	The gap is consistent with training-specific fit, but eight constructed cases do not prove a general capacity cause.

Normal,tied-leaf,missing-flag and changed-data probes

Complete capacity probe	Expected bounded check
Normal supplied paper-tree table	Depth2 fits4/4 but validates 2/4; shallower methodsvalidate 3/4.
Boundary tied leaf near 1 atdepth1	Blocked/clearcount 1/1; declaredclearwins.
Missing near flag for proposed new query	Cannot route the declared binary-input tree; request evidence/review, not invented 0.
Changed evaluation distribution or label policy	A gap may have a different explanation; record the version/coverage change before attributing it to capacity.

Optional complete constrained fitter

```
def majority(rows):
    labels = [row[2] for row in rows]
    return 'blocked' if labels.count('blocked') > labels.count('clear') else 'clear'


def fit_node(rows, remaining_depth, feature=0):
    if not rows:
        raise ValueError('This fixture requires a nonempty node')
    fallback = majority(rows)
    if remaining_depth == 0 or len({row[2] for row in rows}) == 1:
```

```
        return {'label': fallback}
    children = {}
    for value in (0, 1):
        subset = [row for row in rows if row[feature] == value]
        children[value] = (fit_node(subset, remaining_depth - 1, feature + 1)
                           if subset else {'label': fallback})
    return {'feature': feature, 'children': children}


def predict(features, tree):
    if len(features) != 2 or any(type(x) is not int or x not in (0, 1) for x in features):
        raise ValueError('Need both binary flags in near/tall order')
    while 'label' not in tree:
        tree = tree['children'][features[tree['feature']]]
    return tree['label']


# Unique trip IDs are printed in the lesson table; tuples hold near,tall,reference.
training = [(0, 0, 'clear'), (0, 1, 'clear'),
            (1, 0, 'blocked'), (1, 1, 'clear')]
validation = [(0, 0, 'clear'), (0, 1, 'clear'),
              (1, 0, 'clear'), (1, 1, 'blocked')]
# Human-declared question order: near first, then tall; leaf ties choose clear.
for depth in [0, 1, 2]:
    tree = fit_node(training, depth)
    train_correct = sum(predict(row[:2], tree) == row[2] for row in training)
    valid_correct = sum(predict(row[:2], tree) == row[2] for row in validation)
    print(depth, train_correct, len(training), valid_correct, len(validation))
# Caller must supply complete binary training tuples/allowed labels and depth0..2.
# This is a constrained classroom fitter, not a general optimized tree algorithm.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Complexity Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.

1. Write the hypothesis and expected **evidence** before running.

2. **Complexity** Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare two fresh train/validation results

> Keep the chosen candidate and the gap's diagnostic limit separate.

Complete independent results and scope

Model A **training**96%, validation62%; model B **training**88%, validation79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

Use the selected A:training 96%, validation 62%; B:training 88%, validation 79%, on the stated same valid evaluation conditions. Compute training-minus-validation accuracy gaps in percentage points, then choose by validation accuracy. Do not substitute the pictured 85/80or100/60 results or the paper-tree denominators. No causal diagnosis or untouched final result follows from this aggregate comparison alone.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Model A has **training** accuracy 99% and validation accuracy **70%**; B has 88% and 84%. Compute each gap and discuss the **evidence**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's capacity evidence tied to its data

> A training-validation gap is an investigation signal,not a standalone release rule.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

Keep capacity setting, training/validation source versions, metric, denominators, per-case outputs and all development attempts. Compare against a simple baseline and inspect coverage/leakage/label differences before attributing a gap to flexibility. A small gap can reflect two poor methods, a changed gap can reflect data differences, and a larger model can still become the better development candidate.

Freeze before any separate final claim.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

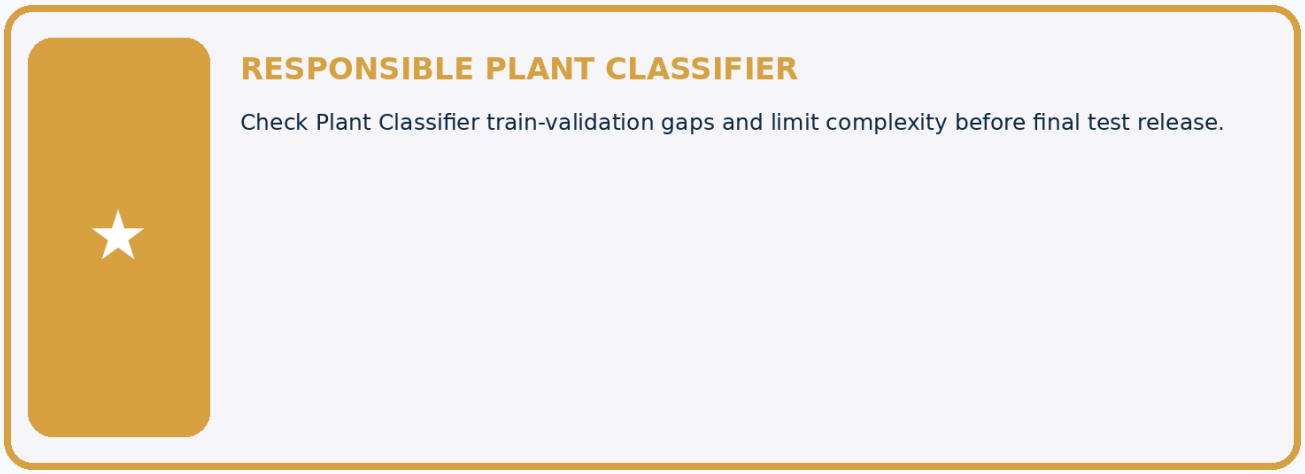
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with the selection rule and a competing explanation

> Explain why capacity,training fit and gap size need distinct evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
overfitting	fit to noise rather than general pattern	
complexity	model capacity to form detailed boundaries	
generalization gap	train score minus unseen score	
regularization	constraint discouraging excessive complexity	

Closer-explanation terms

Word	Meaning
Capacity	Flexibility of a model to represent patterns
Overfitting	Learning details that fail to generalize well
Underfitting	Failing to capture useful structure

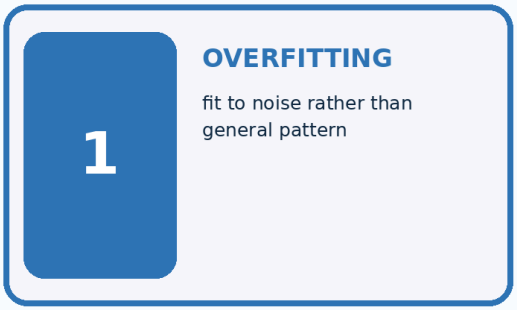
Word	Meaning
Generalization	Performance on relevant unseen examples

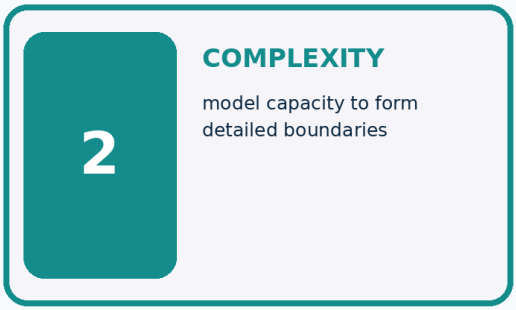
▼ Earlier wording, alternatives and picture transcripts

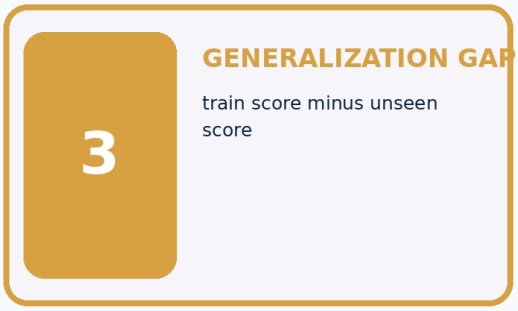
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR The model with the highest **training** accuracy is the best model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **overfitting**, **complexity**, **generalization gap**, **regularization**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The model with the highest **training** accuracy is the best model.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

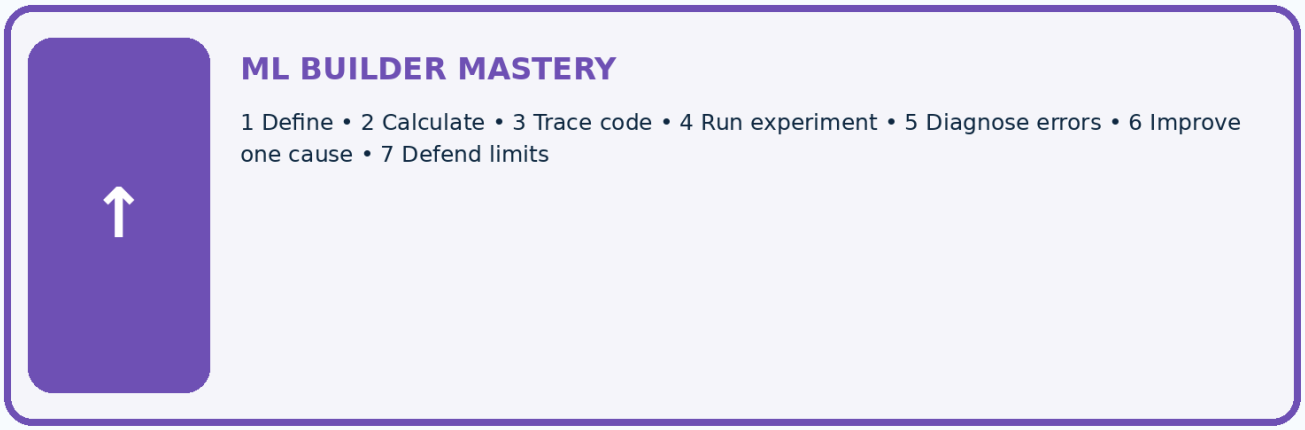
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh96/62versus88/79accuracy comparison for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

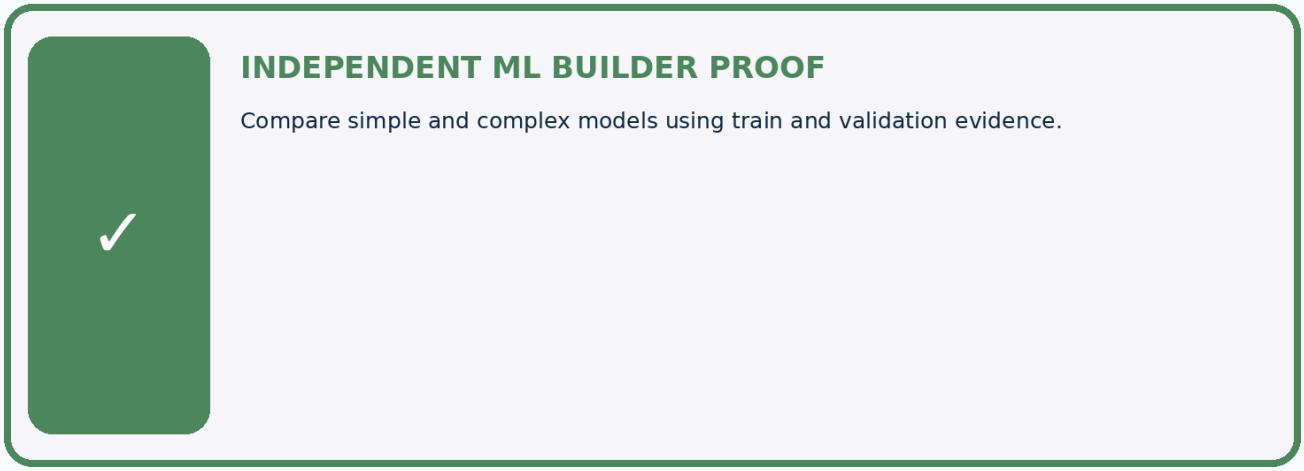
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Compare simple and complex models using train and validation **evidence**.

NEXT BRIDGE Carry the experiment **evidence** into: Learning Curves.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.

Compute gaps and select by validation accuracy.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

<details>
<summary>Check after trying</summary>

A gap40 percentage points; B gap5. B has better shared validation accuracy. A suggests **overfitting**, but investigate distribution differences too.

Evidence or stage	Value or result
A training	100
A validation	60
B training	85
B validation	80

</details>

▼ Earlier wording, alternatives and picture transcripts

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare training and shared validation

A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.

Compute gaps and select by validation accuracy.

A gap40 percentage points; B gap5. B has better shared validation accuracy. A suggests **overfitting**, but investigate distribution differences too.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: See **overfitting** as a gap requiring investigation, not a label based only on model size.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- See **overfitting** as a gap requiring investigation, not a label based only on model size.

- Which labelled step matches this detail? Check **capacity**, **training** choices and data mismatch without tuning on final cases.
- 1. Select development candidate
- 2. Compute gaps
- 3. Investigate large model
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Investigate large model Check **capacity**, **training** choices and data mismatch without tuning on final cases. Read the picture in order.

Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Large’s validation accuracy rises to82% after a development revision. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Its gap is 18 percentage points; it now beats Small on this validation set.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set.

Validation performance improved, but the **training**-validation gap is not a complete release rule.

Repair and keep the record

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

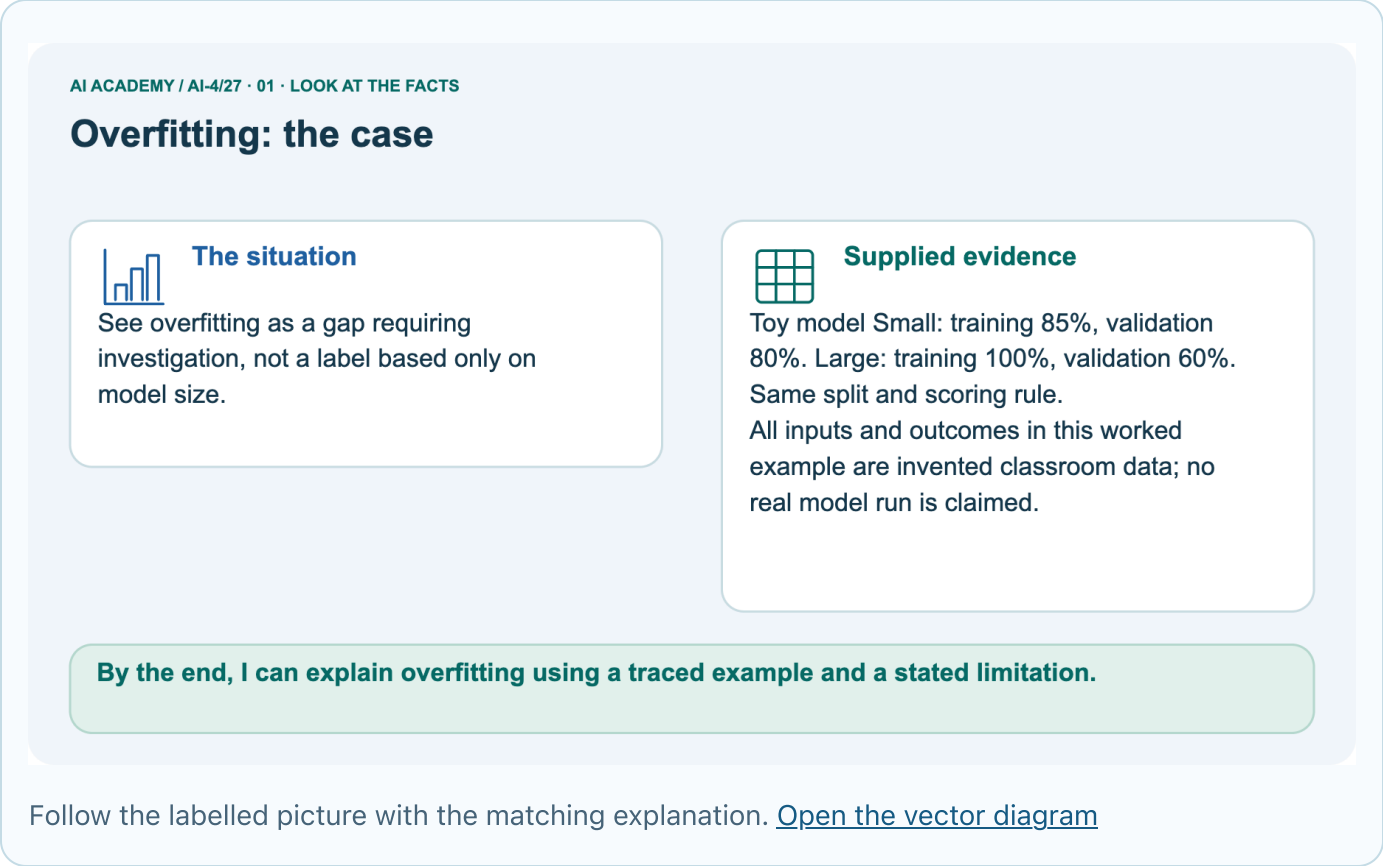
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Overfitting: the case



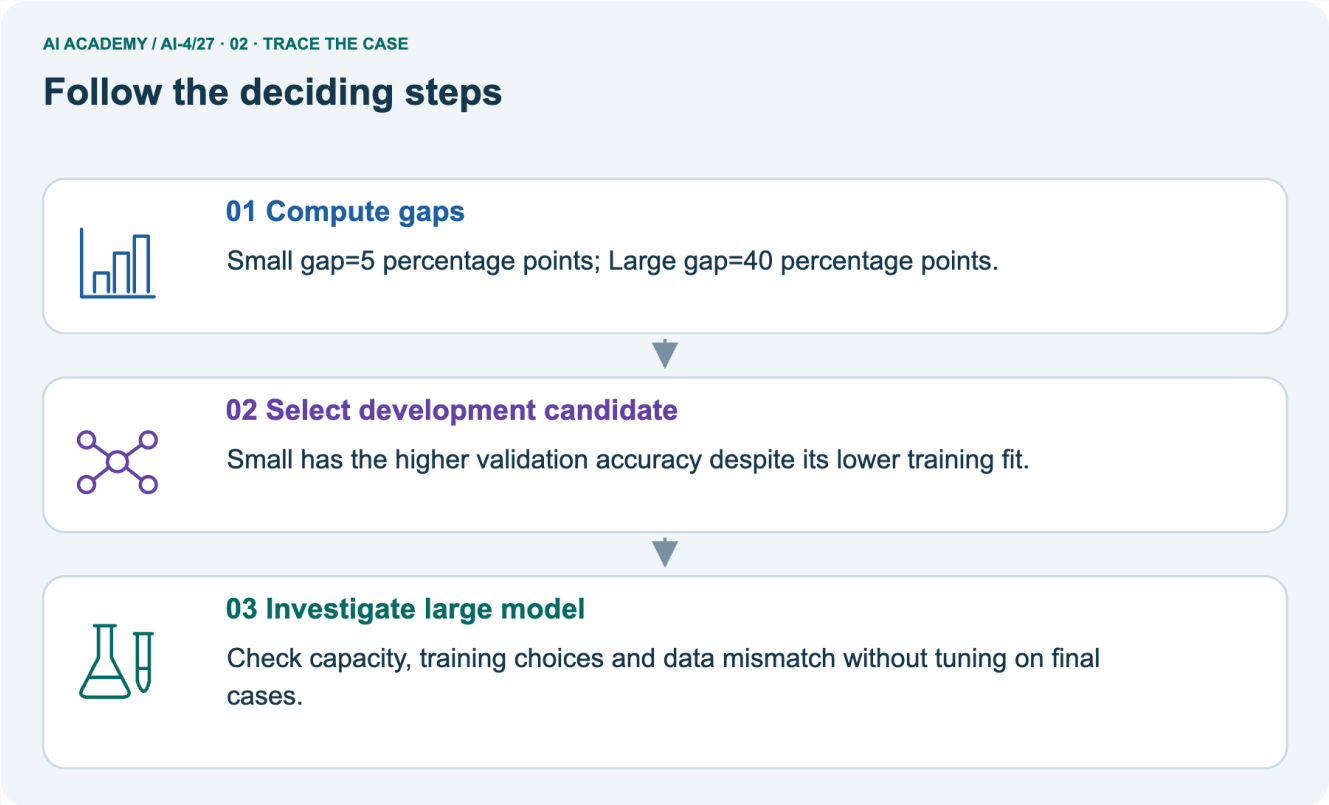
Follow the labelled picture with the matching explanation. Open the vector diagram

See **overfitting** as a gap requiring investigation, not a label based only on model size. Supplied facts: Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

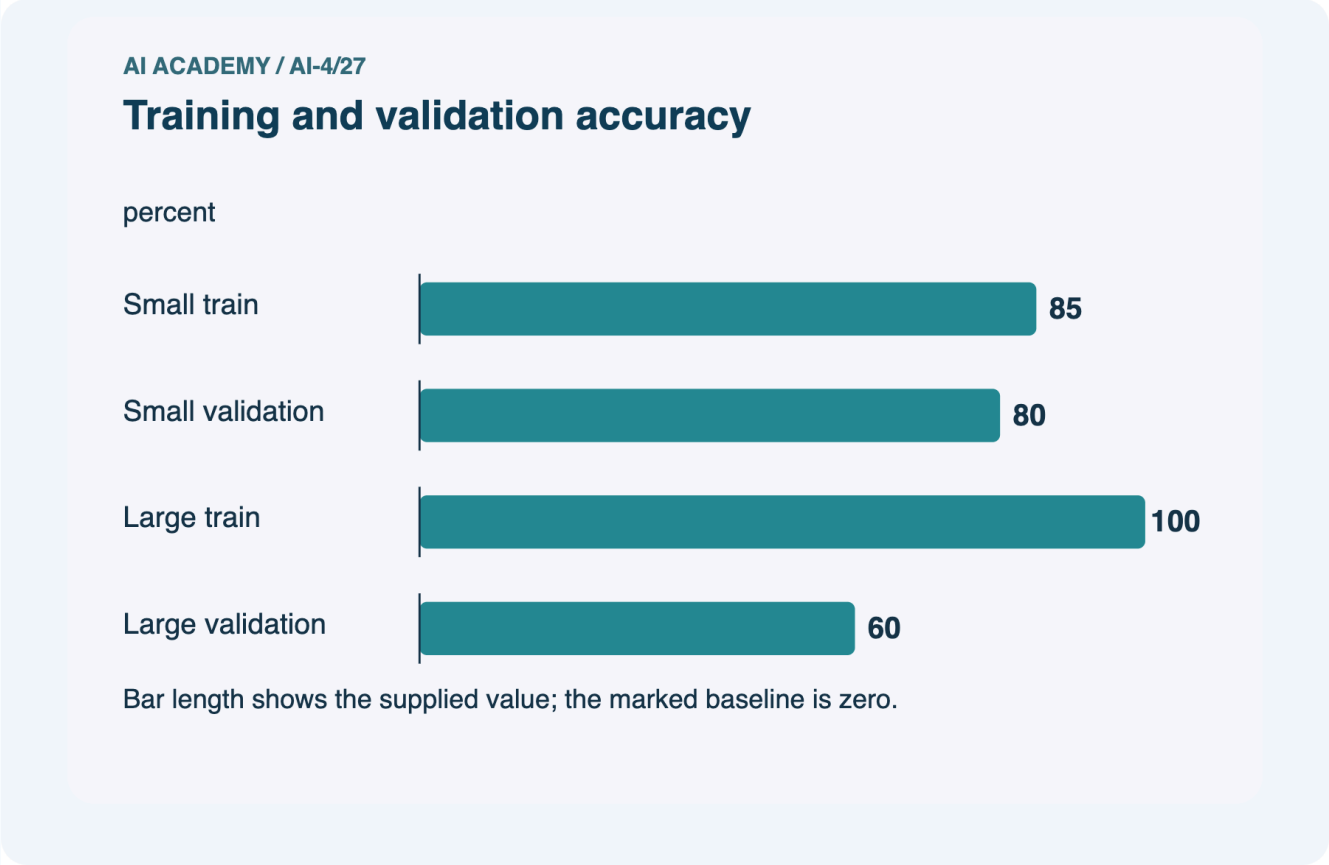


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gaps: Small gap=5 percentage points; Large gap=40 percentage points. → Select development candidate: Small has the higher validation accuracy despite its lower **training** fit. → Investigate large model: Check **capacity**, **training** choices and data mismatch without tuning on final cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Training and validation accuracy

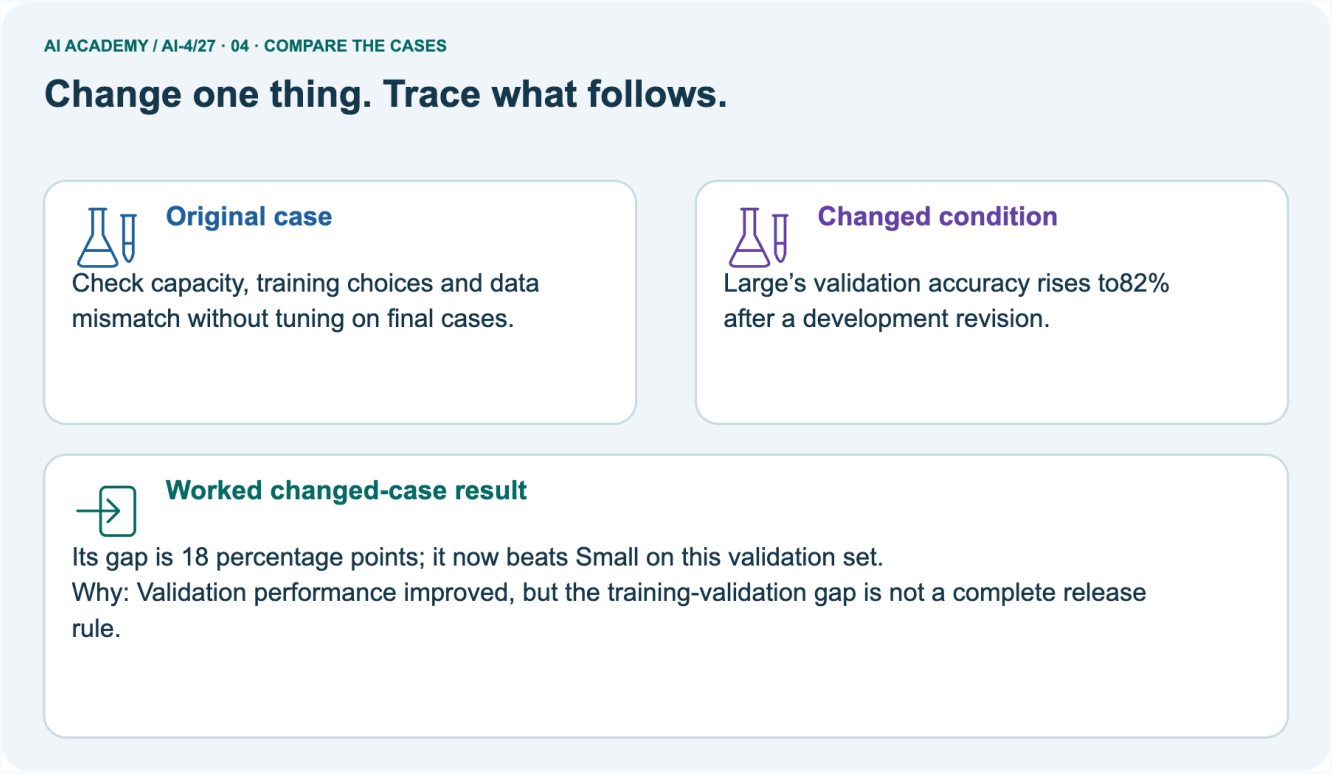


Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation accuracy. Units: percent. Small train: 85. Small validation: 80. Large train: 100. Large validation: 60.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



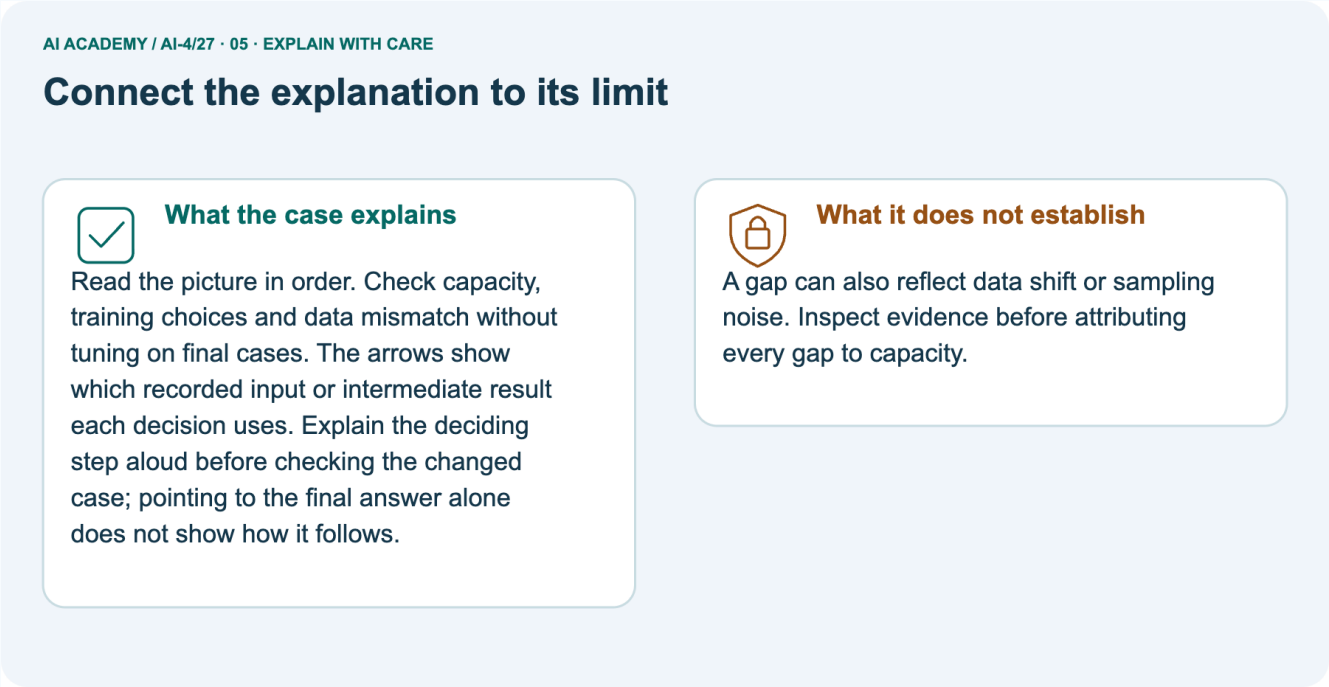
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Check **capacity**, **training** choices and data mismatch without tuning on final cases. Changed condition: Large's validation accuracy rises to82% after a development revision.

Expected result: Its gap is 18 percentage points; it now beats Small on this validation set. Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

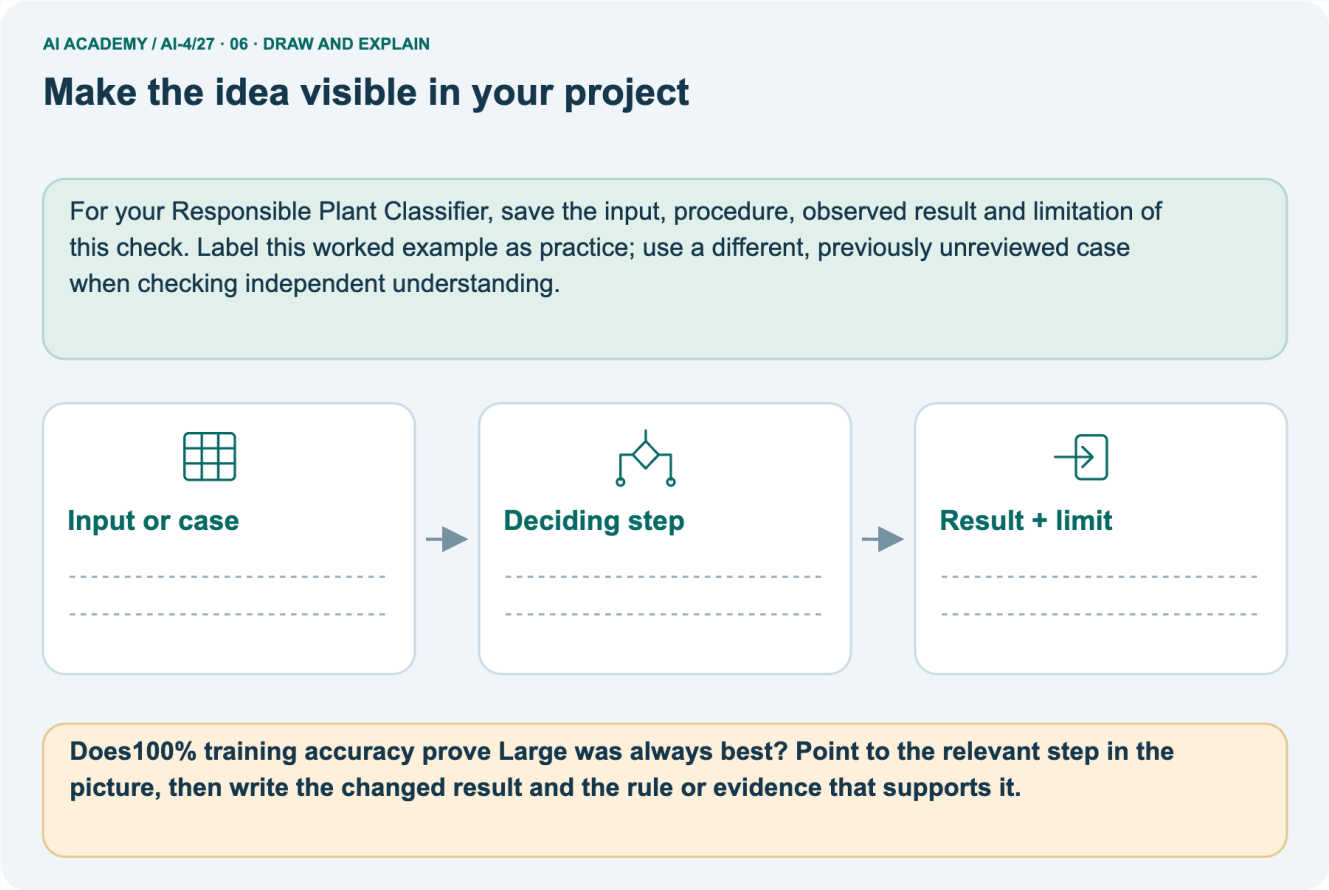
Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A gap can also reflect data shift or sampling noise.

Inspect **evidence** before attributing every gap to **capacity**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Overfitting: the case

02. Follow the deciding steps

03. Training and validation accuracy

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.